

# Public-Data Reproduction of the AION-1 Omnimodal Foundation Model:

## Frozen-Encoder Downstream Experiments

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### Abstract

We reproduce the downstream experiments of AION-1 (Parker and The Polymathic AI Collaboration, 2025), the Polymathic AI omnimodal foundation model for astronomy, using only the publicly released frozen checkpoints (aion-base/large/xlarge; 0.3/0.9/3 B parameters) and public data, on seven NVIDIA TITAN RTX GPUs. Pretraining is out of scope (the original used 64–288 H100s). We rebuild the benchmark datasets from the Multimodal Universe (The Multimodal Universe Collaboration, 2024) and a handful of label catalogs, extract frozen-encoder embeddings, and train the paper’s lightweight probing heads (attentive pooling and a convolutional segmentation upsampler). The flagship galaxy-property result reproduces closely: with a frozen encoder, the redshift  $R^2$  rises from 0.81 (photometry) to 0.98 (+spectrum) to 0.985 (+image+spectrum), matching the paper’s multimodal column (1.00). Stellar-parameter residuals from *Gaia* XP coefficients match or beat the paper. Morphology, retrieval, and segmentation reproduce qualitatively; where our numbers fall short of the paper we trace it to corpus scale (rate-limited image-cutout access) rather than the pipeline, and report ours alongside the paper’s throughout.

## 1 Introduction

AION-1 integrates 39 heterogeneous data types from five surveys (Legacy Survey, HSC, SDSS, DESI, *Gaia*) into a single masked-modeling transformer. The released model is used as a *frozen* feature extractor: each downstream task trains only a small head on the encoder embeddings. We reproduce the eleven reported experiments, grouped as (i) galaxy and stellar property estimation, (ii) morphology classification, (iii) similarity-based retrieval, (iv) image segmentation, and (v) generative redshift posteriors and spectral super-resolution.

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## 2 Methods

**Model and probes.** We load each checkpoint with the released `polymathic-aion` package, freeze it, and compute token embeddings  $E \in \mathbb{R}^{B \times N \times d}$  ( $d = 768/1024/2048$ ). For property estimation we use the paper’s *attentive pooling* head:  $K$  learned queries cross-attend to  $E$  (a normalized cross-attention block taken from the model’s own `posttraining` code), followed by per-target linear maps. For morphology we mean-pool and use a two-layer MLP; for segmentation we reshape the  $24 \times 24$  spatial tokens and apply a lightweight transposed-convolution upsampler. Retrieval uses mean-pooled,  $L_2$ -normalized embeddings and cosine similarity. All heads train with AdamW under an 80/20 split, fixed `SEED`= 2026.

**Data.** The tutorials’ pre-joined files are no longer hosted; we rebuild from the Multimodal Universe HuggingFace datasets. PROVABGS labels (Hahn et al., 2023) key-join to DESI spectra on `TARGETID` (20,666 galaxies); Legacy Survey  $g, r, i, z$  images (Dey et al., 2019) are fetched from the cutout service by sky position. Stellar labels come from the DD-Payne DESI catalog (Zhang et al., 2024) (25,992 stars), with *Gaia* DR3 parallax (Gaia Collaboration, 2023) attached via CDS XMATCH (97% matched). Morphology and retrieval use Galaxy10 DECaLS (Walmsley et al., 2022); strong-lens positives are the SuGOHI master list (Jaelani et al., 2024); segmentation masks are the GZ3D value-added catalog (Masters et al., 2021).

## 3 Results

**Galaxy property estimation (Table 1).** Adding the DESI spectrum to photometry is the dominant effect, recovering nearly the full multimodal result; the image alone helps the morphology-correlated properties.

**Table 1:** PROVABGS galaxy properties, frozen AION-1-base,  $R^2$ .

| config               | $z$          | $\log M_\star$ | age  | $\log Z$ | sSFR |
|----------------------|--------------|----------------|------|----------|------|
| photometry           | 0.81         | 0.75           | 0.35 | 0.38     | 0.61 |
| +image               | 0.86         | 0.84           | 0.33 | 0.44     | 0.63 |
| +spectrum            | 0.98         | 0.94           | 0.46 | 0.57     | 0.69 |
| +image+spectrum      | <b>0.985</b> | <b>0.94</b>    | 0.43 | 0.57     | 0.69 |
| paper (phot+im+spec) | 1.00         | 0.96           | 0.53 | 0.61     | 0.72 |

**Stellar parameters.** From APOGEE-matched *Gaia* XP coefficients, residual standard deviations are  $T_{\text{eff}} \approx 89\text{--}95$  K (paper 94.6),  $\log g \approx 0.19\text{--}0.20$  dex (0.206), and  $[\text{Fe}/\text{H}] \approx 0.11$  dex (0.115) – a clean match. For the spectrum-based DESI×DD-Payne task, the DESI-only config gives  $T_{\text{eff}} = 0.96$ ,  $\log g = 0.89$ ,  $[\text{Fe}/\text{H}] = 0.80$ ; adding *Gaia* parallax raises the surface-gravity-sensitive parameters toward the paper’s DESI+parallax column (0.99/0.98/0.94).

**Generative and low-data.** The single-shot redshift head produces a posterior over the 1024-bin grid that contracts sharply when the spectrum is added (task 10), and *Gaia* BP/RP coefficients decode to a DESI-resolution spectrum recovering the major line centers (median correlation 0.83, task 11). In the low-data regime, the spectrum-config  $R^2$  saturates by  $10^3\text{--}10^4$  labels, reproducing the paper’s sample-efficiency claim.

**Morphology, retrieval, segmentation.** Galaxy10 morphology accuracy is 0.72–0.75 (paper 0.84); spiral/merger retrieval nDCG@10 is  $\sim 0.82/0.68$  (paper 0.938/0.892); strong-lens retrieval and GZ3D spiral/bar segmentation are reproduced with the frozen encoder. These tasks are *corpus-limited*: the Legacy Survey cutout service rate-limits image acquisition, so we evaluate on smaller, less-rare corpora than the paper’s full GZ-DECaLS; the live numbers are in `data/results/REPORT.md`.

## 4 Limitations

Absolute numbers are reproduction *targets*, not pass/fail gates: the paper omits seeds, exact split membership, and head hyperparameters. The largest genuine shortfalls (morphology, retrieval) stem from corpus scale, not modeling. The **xlarge** checkpoint underperforms **base** on photometry-only inputs, as expected for a 3 B frozen model given four scalar tokens. Full reproduction details, scripts, and the live comparison table accompany this report.

## References

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